



Shah Md. Tanvir Siddiquee

**Assistant Professor, Department of CSE
Daffodil International University**

 +8801712560297  tanvir.cse@diu.edu.bd

 1st Floor, Room No. 111, Daffodil Tower5

 <https://sites.google.com/diu.edu.bd/tanvirsiddiquee/>

Elimination of Left Factoring (LF)

Grammar

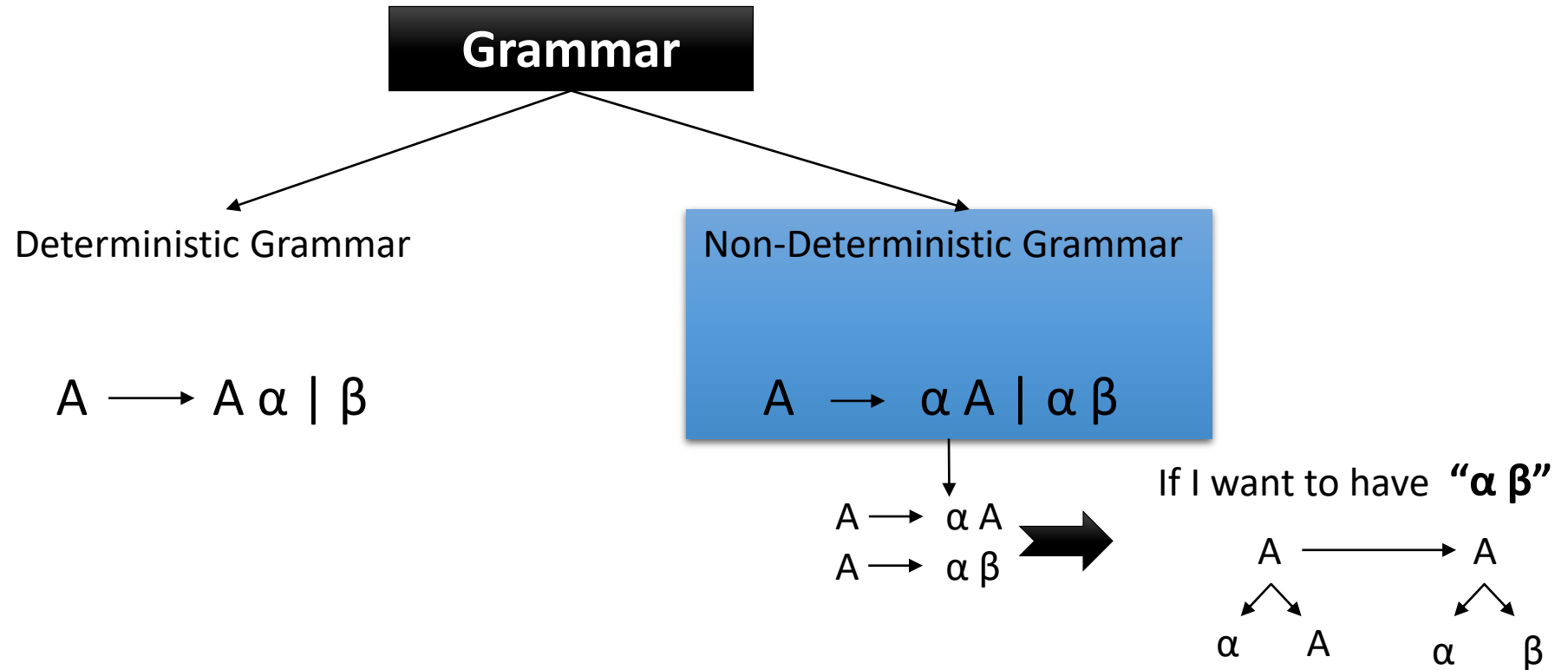
```
graph TD; Grammar[Grammar] --> Deterministic[Deterministic Grammar]; Grammar --> NonDeterministic[Non-Deterministic Grammar]; Deterministic --> Rule1["A → A α | β"]; NonDeterministic --> Rule2["A → α A | α β"];
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Deterministic Grammar

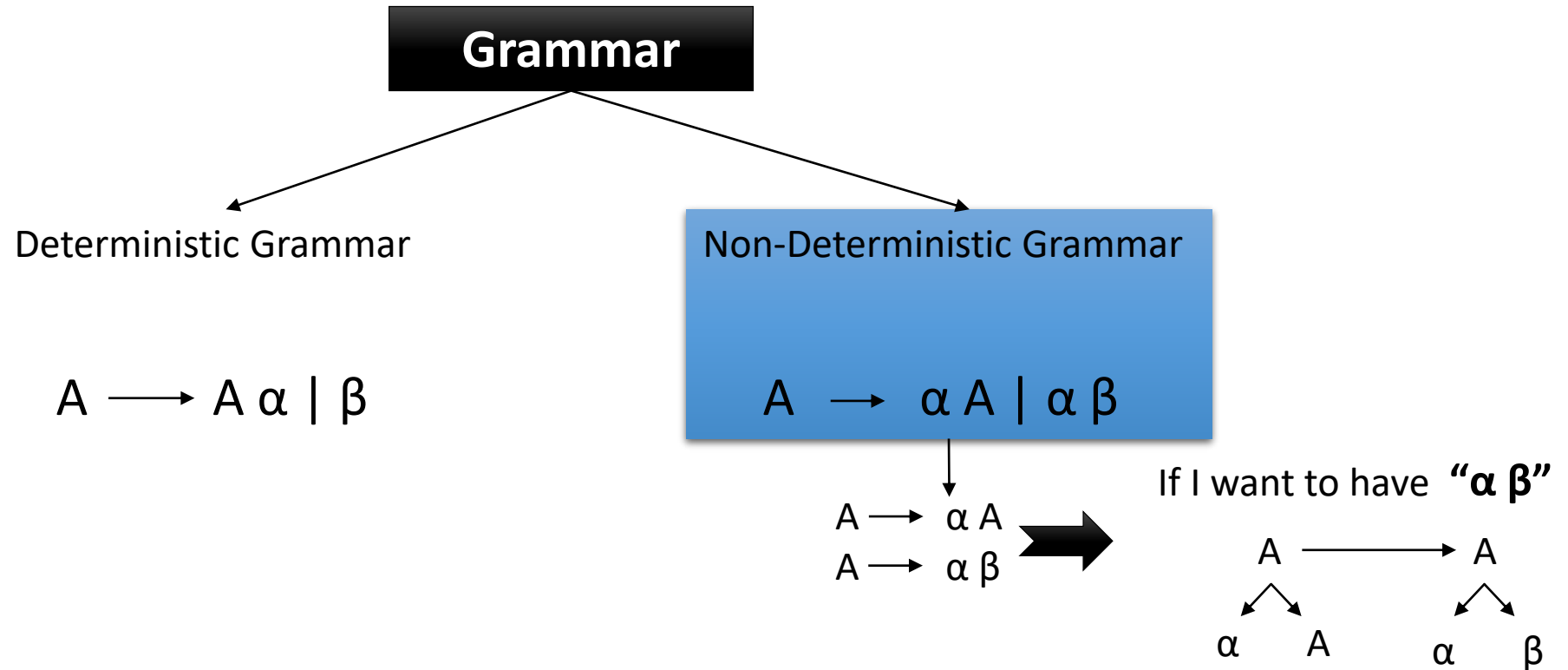
$$A \longrightarrow A \alpha \mid \beta$$

Non-Deterministic Grammar

$$A \longrightarrow \alpha A \mid \alpha \beta$$

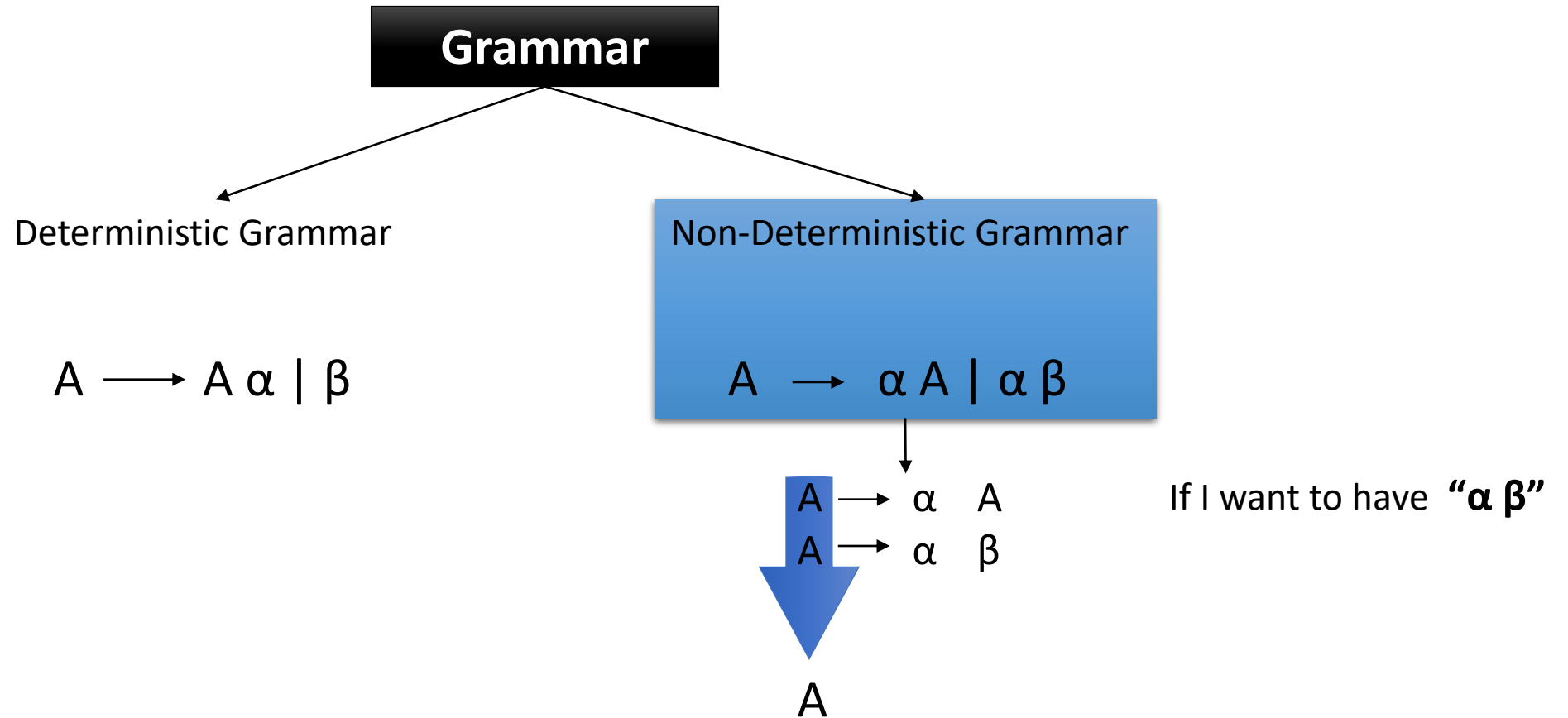


As we can derive both αA & $\alpha \beta$ from the grammar, both have same prefix (α), so, the grammar is non-deterministic.



As we can derive both αA & $\alpha \beta$ from the grammar, both have same prefix (α), so, the grammar is non-deterministic.

To make this grammar deterministic, we can apply left factoring.



Grammar

Deterministic Grammar

$$A \longrightarrow A \alpha \mid \beta$$

Non-Deterministic Grammar

$$A \longrightarrow \alpha A \mid \alpha \beta$$

$$\begin{array}{l} A \longrightarrow \alpha A \\ A \longrightarrow \alpha \beta \end{array}$$

$$A \longrightarrow \alpha$$

If I want to have “ $\alpha \beta$ ”

Grammar

Deterministic Grammar

$$A \longrightarrow A \alpha \mid \beta$$

Non-Deterministic Grammar

$$A \longrightarrow \alpha A \mid \alpha \beta$$

$$\begin{array}{l} A \longrightarrow \alpha \quad A \\ A \longrightarrow \alpha \quad \beta \\ \downarrow \\ A \longrightarrow \alpha \quad A' \end{array}$$

If I want to have “ $\alpha \beta$ ”

Grammar

Deterministic Grammar

$$A \longrightarrow A \alpha \mid \beta$$

Non-Deterministic Grammar

$$A \longrightarrow \alpha A \mid \alpha \beta$$

$$\begin{array}{l} A \longrightarrow \alpha A \\ A \longrightarrow \alpha \beta \end{array}$$

If I want to have “ $\alpha \beta$ ”

$$\begin{array}{l} A \longrightarrow \alpha A' \\ \downarrow \\ A' \longrightarrow A \mid \beta \end{array}$$

Grammar

Deterministic Grammar

$$A \longrightarrow A \alpha \mid \beta$$

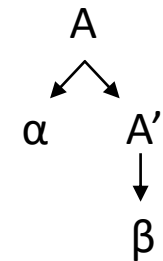
Non-Deterministic Grammar

$$A \longrightarrow \alpha A \mid \alpha \beta$$

$$\begin{array}{l} A \longrightarrow \alpha A \\ A \longrightarrow \alpha \beta \end{array}$$

If I want to have “ $\alpha \beta$ ”

$$\begin{array}{l} A \longrightarrow \alpha A' \\ A' \longrightarrow A \mid \beta \end{array}$$



Exercise:

$$\begin{aligned} S &\rightarrow iEtS \\ &\quad | iEtSeS \\ &\quad | a \\ E &\rightarrow b \end{aligned}$$

Solution:

$$\begin{aligned} S &\rightarrow iEtS \\ &\quad | iEtSeS \\ &\quad | a \\ E &\rightarrow b \end{aligned}$$

$$A \rightarrow \alpha A \mid \alpha \beta$$

$$A \rightarrow \alpha A'$$

$$A' \rightarrow A \mid \beta$$

Exercise:

$$\begin{aligned} S &\rightarrow iEtS \\ &\quad | iEtSeS \\ &\quad | a \\ E &\rightarrow b \end{aligned}$$

$$A \rightarrow \alpha A \mid \alpha \beta$$

$$A \rightarrow \alpha A'$$

$$A' \rightarrow A \mid \beta$$

Solution:

$$\begin{aligned} S &\rightarrow iEtS \\ &\quad | iEtSeS \\ &\quad | a \\ E &\rightarrow b \end{aligned}$$

Left Factoring

$$\begin{aligned} S &\rightarrow iEtS \ S' \mid a \\ S' &\rightarrow \epsilon \mid eS \\ E &\rightarrow b \end{aligned}$$

Grammar is converted to deterministic.

Exercises for practice:

$$\begin{array}{l} 1 \quad A \longrightarrow aAB \mid aA \\ \quad B \longrightarrow bB \mid b \end{array}$$

$$2 \quad E \longrightarrow T + E \mid T \in$$

$$3 \quad S \longrightarrow bSSaaS \mid bSSaSb \mid bSb \mid a$$

THANK YOU